

 View source on GitHub (https://github.com/tensorflow/tensorflow/blob/v2.16.1/tensorflow/python/ops/array_ops.py#L4254-L4327)

Removes dimensions of size 1 from the shape of a tensor.

```
tf.squeeze(  
    input, axis=None, name=None  
)
```

Used in the notebooks

Used in the guide	Used in the tutorials
• Logistic regression for binary classification with Core APIs (https://www.tensorflow.org/guide/core/logistic_regression_core) • Multilayer perceptrons for digit recognition with Core APIs (https://www.tensorflow.org/guide/core/mlp_core) • Quickstart for the TensorFlow Core APIs (https://www.tensorflow.org/guide/core/quickstart_core) • Unicode strings (https://www.tensorflow.org/text/guide/unicode)	• Learned data compression (https://www.tensorflow.org/tutorials/generative/data_compression) • Generate music with an RNN (https://www.tensorflow.org/tutorials/audio/music_generation) • Simple audio recognition: Recognizing keywords (https://www.tensorflow.org/tutorials/audio/simple_audio) • Intro to Autoencoders (https://www.tensorflow.org/tutorials/generative/autoencoder) • Transfer learning with YAMNet for environmental sound classification (https://www.tensorflow.org/tutorials/audio/transfer_learning_audio)

Given a tensor `input`, this operation returns a tensor of the same type with all dimensions of size 1 removed. If you don't want to remove all size 1 dimensions, you can remove specific size 1 dimensions by specifying `axis`.

For example:

```
# 't' is a tensor of shape [1, 2, 1, 3, 1, 1]  
tf.shape(tf.squeeze(t))  # [2, 3]
```

Or, to remove specific size 1 dimensions:

```
# 't' is a tensor of shape [1, 2, 1, 3, 1, 1]  
tf.shape(tf.squeeze(t, [2, 4]))  # [1, 2, 3, 1]
```

Unlike the older op `tf.compat.v1.squeeze` (https://www.tensorflow.org/api_docs/python/tf/compat/v1/squeeze), this op does not accept a deprecated `squeeze_dims` argument.

★ **Note:** if `input` is a `tf.RaggedTensor` (https://www.tensorflow.org/api_docs/python/tf/RaggedTensor), then this operation takes $O(N)$ time, where `N` is the number of elements in the squeezed dimensions.

★ **Note:** If `squeeze` is performed on dimensions of unknown sizes, then the returned Tensor will be of unknown shape. A common situation is when the first (batch) dimension is of size `None`, `tf.squeeze` (https://www.tensorflow.org/api_docs/python/tf/squeeze) returns `<unknown>` shape which may be a surprise. Specify the `axis=` argument to get the expected result, as illustrated in the following example:

```
@tf.function  
def func(x):  
    print('x.shape:', x.shape)  
    known_axes = [i for i, size in enumerate(x.shape) if size == 1]  
    y = tf.squeeze(x, axis=known_axes)
```

```
print('shape of tf.squeeze(x, axis=known_axes):', y.shape)
y = tf.squeeze(x)
print('shape of tf.squeeze(x):', y.shape)
return 0

_ = func.get_concrete_function(tf.TensorSpec([None, 1, 2], dtype=tf.int32))
# Output is.
# x.shape: (None, 1, 2)
# shape of tf.squeeze(x, axis=known_axes): (None, 2)
# shape of tf.squeeze(x): <unknown>
```

Args	
input	A Tensor . The input to squeeze.
axis	An optional list of ints . Defaults to <code>[]</code> . If specified, only squeezes the dimensions listed. The dimension index starts at 0. It is an error to squeeze a dimension that is not 1. Must be in the range <code>[-rank(input), rank(input))</code> . Must be specified if input is a RaggedTensor .
name	A name for the operation (optional).
Returns	
A Tensor . Has the same type as input . Contains the same data as input , but has one or more dimensions of size 1 removed.	
Raises	
ValueError	The input cannot be converted to a tensor, or the specified axis cannot be squeezed.

Except as otherwise noted, the content of this page is licensed under the [Creative Commons Attribution 4.0 License](https://creativecommons.org/licenses/by/4.0/) (<https://creativecommons.org/licenses/by/4.0/>), and code samples are licensed under the [Apache 2.0 License](https://www.apache.org/licenses/LICENSE-2.0) (<https://www.apache.org/licenses/LICENSE-2.0>). For details, see the [Google Developers Site Policies](https://developers.google.com/site-policies) (<https://developers.google.com/site-policies>). Java is a registered trademark of Oracle and/or its affiliates. Some content is licensed under the [numpy license](https://numpy.org/doc/stable/license.html) (<https://numpy.org/doc/stable/license.html>).